

SQL DDL Script Writing Exercises

Exercise 1 — Creating a Database and Table

Create a database named:

`InventoryDB`

Write the statement so that the database is created only if it does not already exist.

Make `InventoryDB` the default database.

Then create a table named `Product` with the following columns. Do not create the tables if it already exists.

Column	Data Type	Constraint
ProductID	Number	Primary Key
ProductName	String	Cannot be NULL
Price	Money	6 total digits, 2 digits after decimal point
Quantity	Number	

Then create a table named `Category` with the following columns:

Column	Data Type	Constraint
CategoryID	Number	Primary Key
CategoryName	String	Must be Unique

Exercise 2 — Relate the Tables using a Foreign Key

Create a foreign key so that:

`Product.CategoryID`

references:

`Category.CategoryID`

Make sure the tables are created in the correct order. This follows the same foreign-key concept demonstrated in the presentation without reusing the classroom entities.

Exercise 3 — Using **ALTER TABLE** to Add Columns

Assume the following table already exists (copy/paste this in your script):

```
CREATE TABLE Customer (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(100) NOT NULL  
);
```

Add the following columns to the Customer table (do not recreate the table):

Email	String of length 100
Phone	String of length 20
City	String of length 30

Exercise 4 — Modifying, Renaming, and Dropping Columns

Assume the following table already exists (copy/paste this in your script):

```
CREATE TABLE Employee (  
    EmployeeID INT PRIMARY KEY,  
    EmployeeName VARCHAR(50) NOT NULL,  
    JobTitle VARCHAR(30),  
    Extension INT  
);
```

Using **MySQL**, write statements to:

1. Change `EmployeeName` to length of 100.
 2. Rename `JobTitle` to `Position`.
 3. Remove the `Extension` column.
-

Exercise 5 — Adding a Constraint with **ALTER TABLE**

Assume this MySQL table already exists (copy/paste this in your script):

```
CREATE TABLE Customer (  
    CustomerID INT PRIMARY KEY,  
    CustomerName VARCHAR(75) NOT NULL,  
    Email VARCHAR(100)  
);
```

Write an **ALTER TABLE** statement that makes `Email` unique.

Name the constraint:

```
uq_customer_email
```

Exercise 6 — Using DROP

Suppose the following two tables exist (copy/paste this in your script):

```
CREATE TABLE Category (  
    CategoryID INT PRIMARY KEY,  
    CategoryName VARCHAR(50) UNIQUE  
);  
  
CREATE TABLE Product (  
    ProductID INT PRIMARY KEY,  
    ProductName VARCHAR(50) NOT NULL,  
    Quantity INT,  
    CategoryID INT,  
    FOREIGN KEY (CategoryID)  
        REFERENCES Category(CategoryID)  
);
```

Product contains a foreign key referencing Category.

Write the SQL statements needed to remove both tables.
